

N

Large Scale Recommendation

Personalizing the Screen with SVD

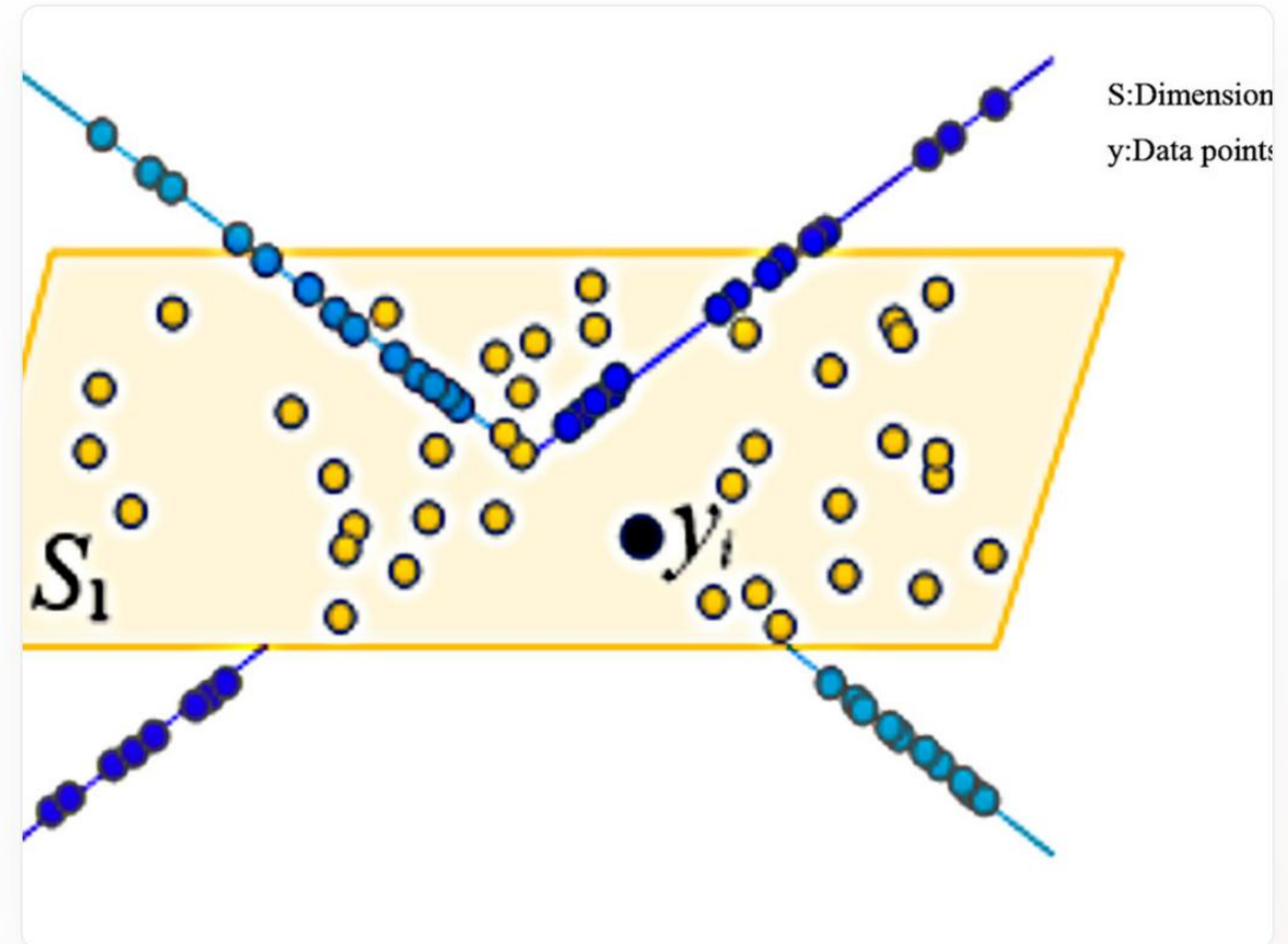
SUBMITTED BY: NIMAI JAIN (24123026)

Navigating the Data Void

The Sparsity Problem

When you look at the raw Netflix Prize dataset, it's mostly empty space. With millions of potential intersections between users and movies, the resulting matrix is over 99% hollow.

Simple statistical models can't "connect the dots" in this void. We need an engine that uncovers the hidden thematic DNA—the latent factors—that actually drive user preferences.



Engineering a Reliable Pipeline



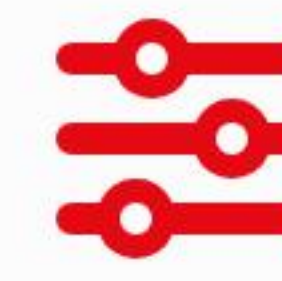
Smart Pruning

We cut the noise by isolating active profiles: users with 80+ ratings and movies with 150+ reviews. This densification sped up training by over 70%.



Leak-Proof Splitting

Random splits are flawed for time-series data. We enforced a strict chronological split—training on the past to predict the genuine future.



Optimization

We tuned the system using Stochastic Gradient Descent (SGD), finding the optimal balance between learning speed and accuracy.

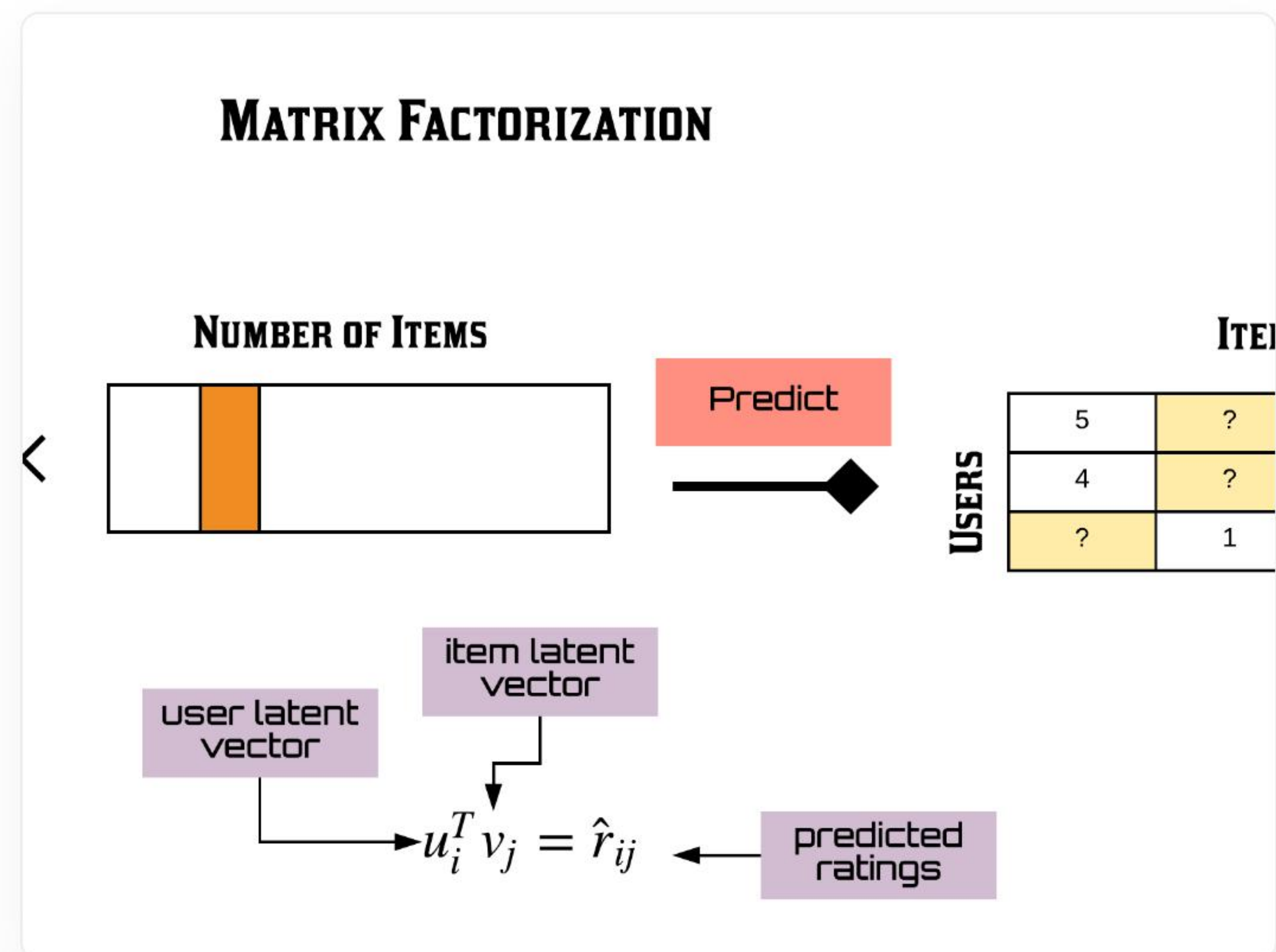
Decomposing Taste into Dimensions

SVD Latent Factors

Our engine maps users and items into a shared space of 30 hidden dimensions. The prediction combines global averages with personal vector alignment:

$$\hat{y}_{u,i} = \mu + b_u + b_i + P_u \cdot Q_i^T$$

By optimizing these latent vectors, the model learns complex thematic relationships without needing manual genre tags.



Performance: Accuracy Comparison

SVD (Latent Model)

RMSE: 0.9699

KNN (Item Baseline)

RMSE: 1.0542

Lower RMSE indicates better accuracy. SVD smooths over empty data regions that trip up traditional neighborhood-based models.

Ranking Quality: Precision at Top-10

53.8%

MAP @ 10 SCORE

Personalizing the Feed

Predicting stars is only half the battle. A useful UI needs to rank relevant items at the very top. SVD achieved a 53.8% MAP@10 score, outperforming the baseline by 12 points.

By balancing item popularity with personalized vector dot products, we avoid the "blockbuster trap" and surface genuine niche matches for the user.

Output Showcase: Sample User Feed

Personalized Top-5 Recommendations for **User Profile 1482031** (Crime & Sci-Fi Enthusiast):

Rank	Movie ID	Predicted Score	Thematic Affinity
1	285	4.85	Classic Neo-Noir Thriller
2	1243	4.62	Sci-Fi Psychological Drama
3	872	4.51	Suspense Crime Detective
4	3144	4.44	Dystopian Cyberpunk Noir
5	1798	4.38	Independent Crime Drama

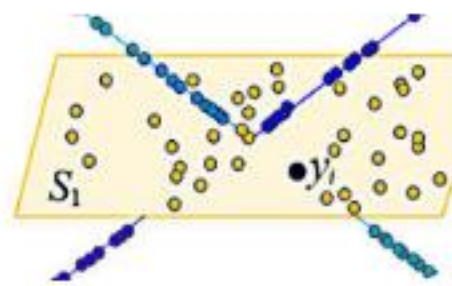
The model successfully identified subtle crime and sci-fi genre patterns without explicit metadata tags.

Image Sources



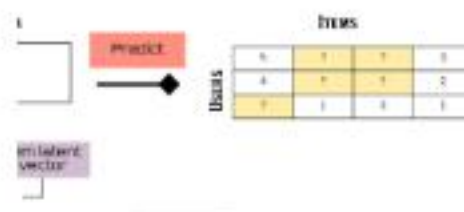
https://static.vecteezy.com/system/resources/previews/020/335/987/non_2x/netflix-logo-netflix-icon-free-free-vector.jpg

Source: www.vecteezy.com



<https://dfzljdn9uc3pi.cloudfront.net/2023/cs-1447/1/fig-1-2x.jpg>

Source: peerj.com



<https://towardsdatascience.com/wp-content/uploads/2020/02/1b4M7o7W8bfRRxdMxtFoVBQ.png>

Source: towardsdatascience.com